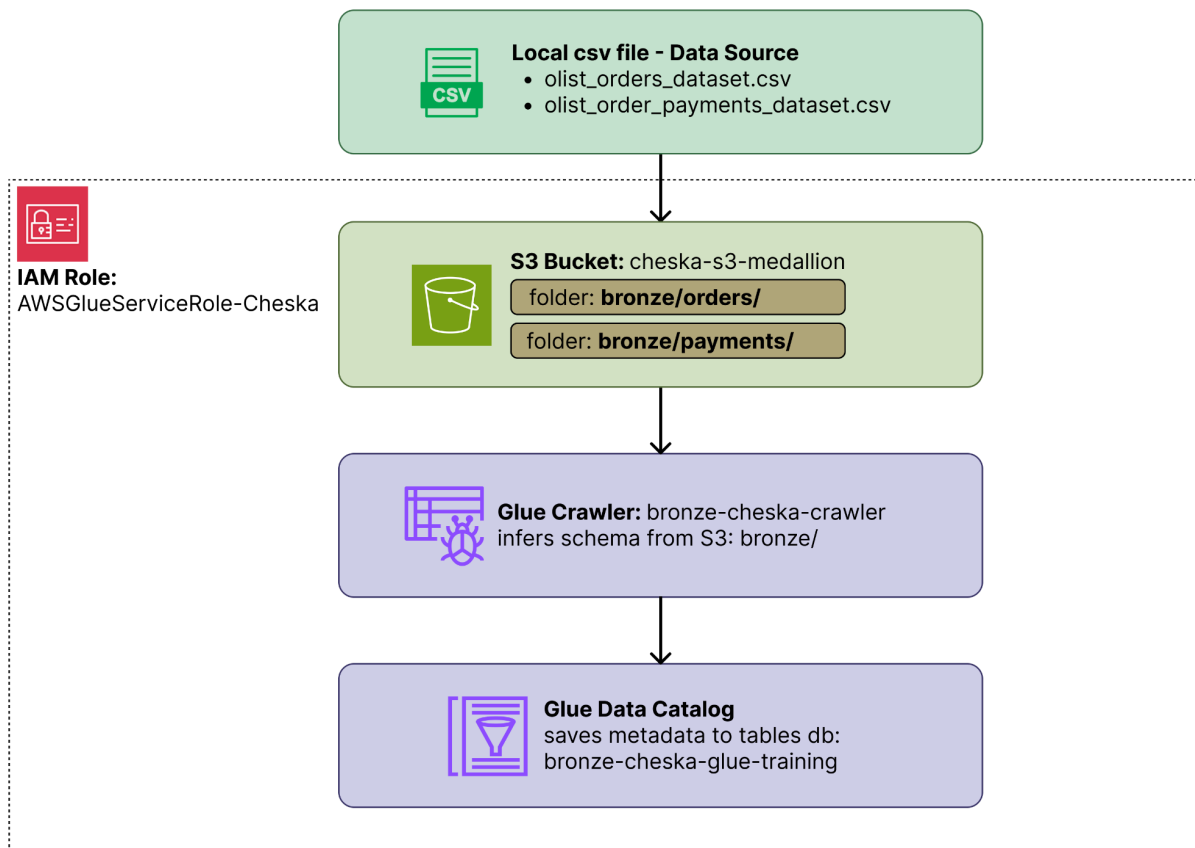


S3 Medallion & Crawler Workflow

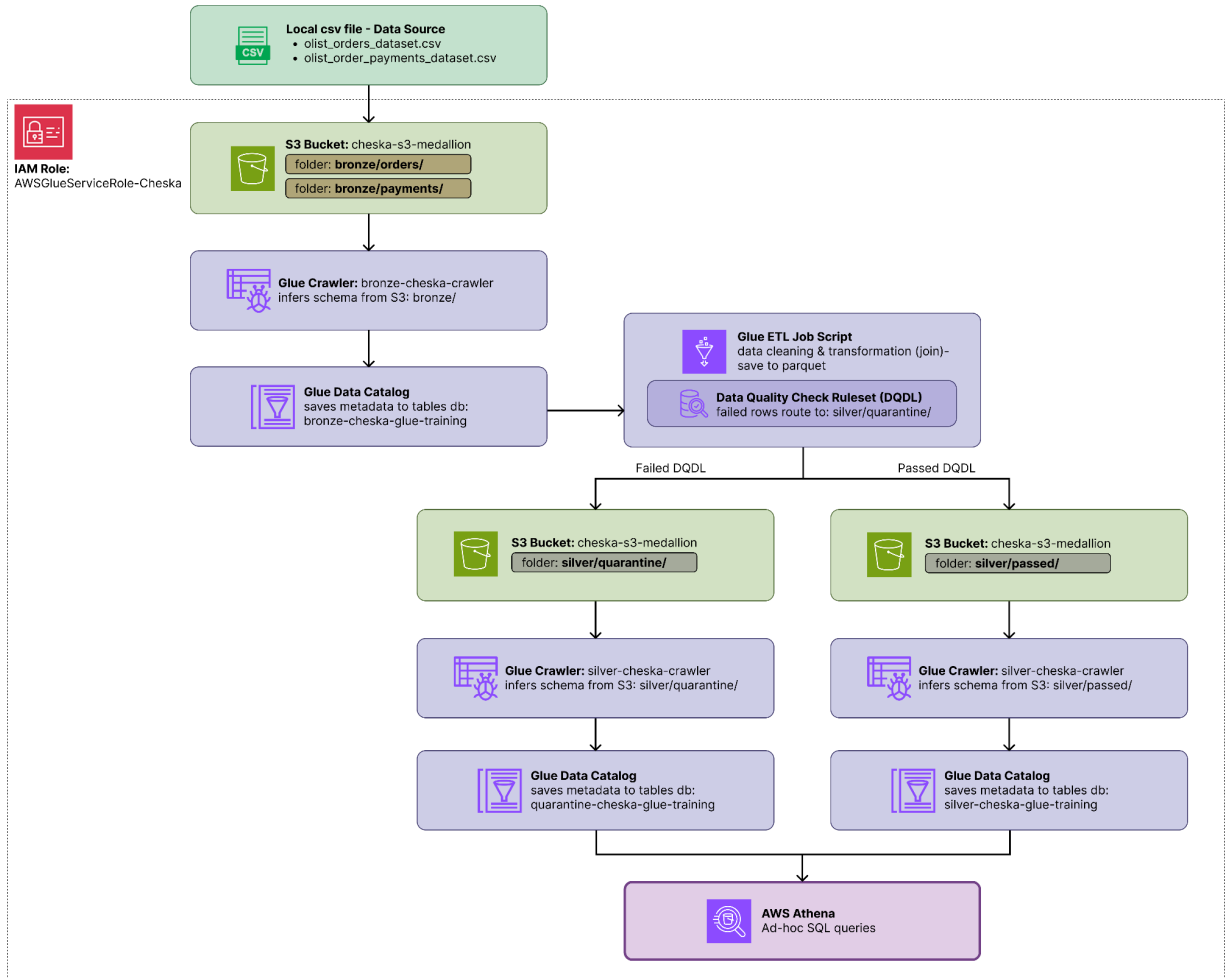
Architecture Diagram:



This architecture diagram provided shows the standard creation and flow of S3 and crawler. From data ingestion through uploading the local files to the S3 bronze folder, using Glue Crawlers to get the metadata of the raw dataset to add into the AWS Data Catalog for easier access and ready for usage on other services such as Athena etc.

Glue ETL with DQDL Workflow

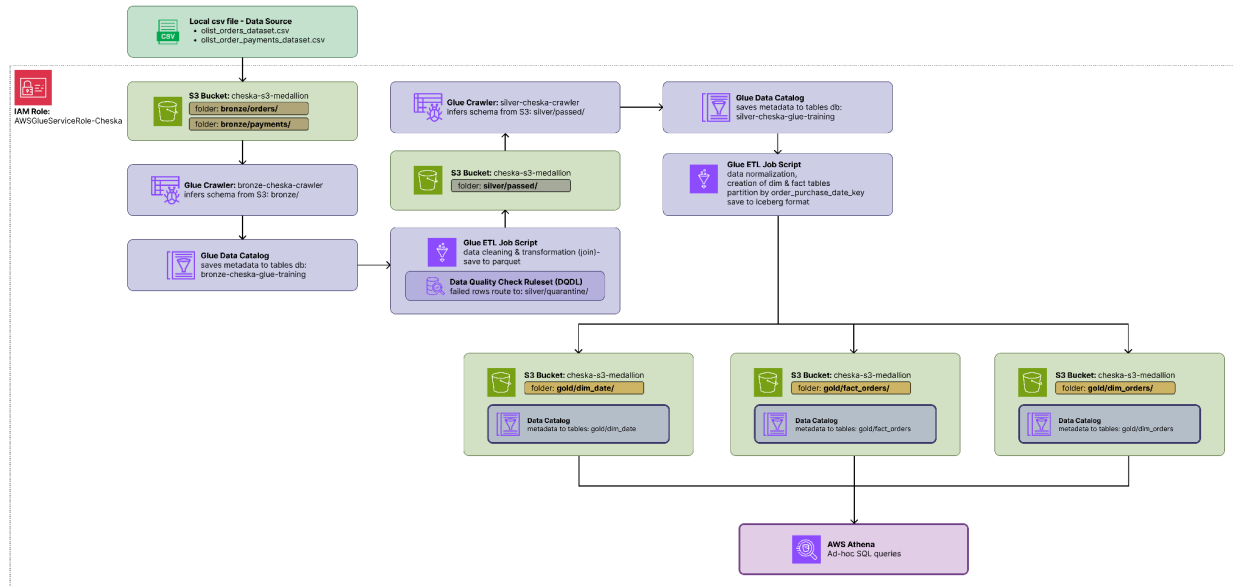
Architecture Diagram:



The diagram above shows the flow from ingestion of the local data into the bronze layer, running a glue crawler to infer the schema and to create a database for data catalog. A glue job ETL script created to perform data cleaning and data evaluation by checking the ruleset made. Once the data has been evaluated, the script will create two flows for the passed data and failed data that will be both saved as parquet files for faster processing. Failed data will be quarantined which will be saved in the S3 `silver/quarantined/` folder and passed data will perform the next transformation in the script, joining the two datasets. Once the inner join is completed, it will be saved in the S3 `silver/passed/` folder. To double check the passed and quarantined data, crawlers were created for both passed and quarantined so that it can be queried by AWS Athena.

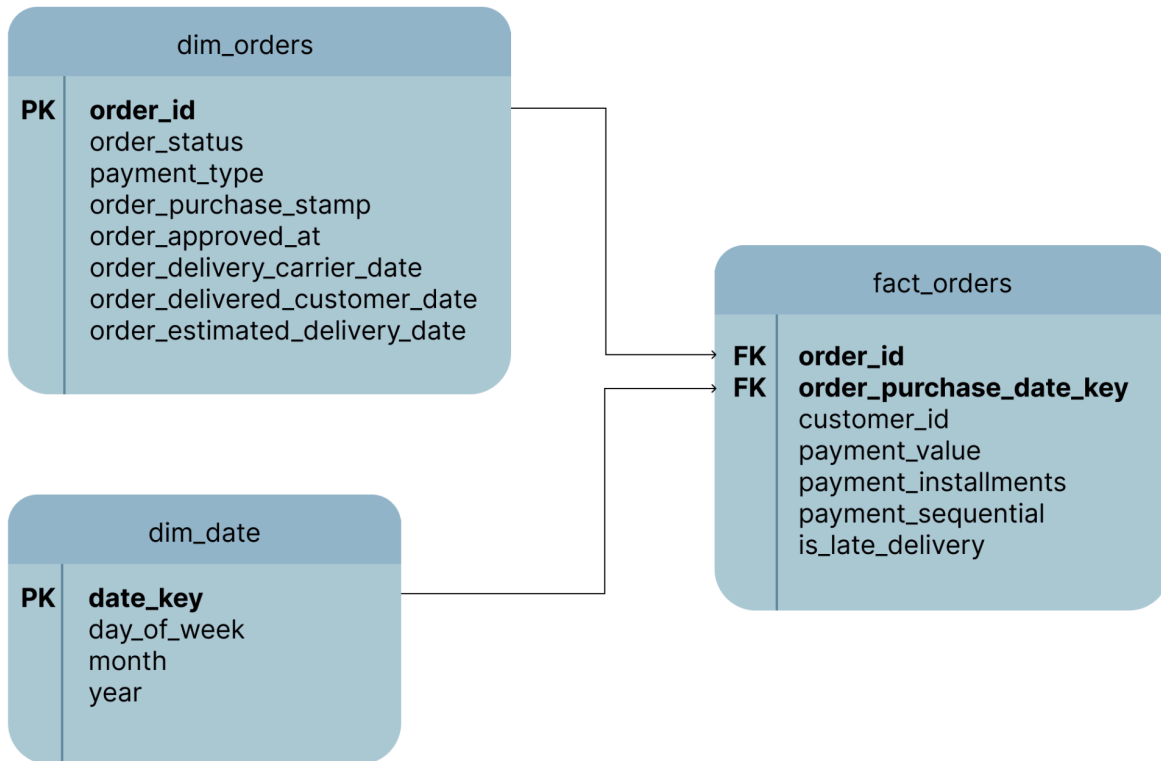
Glue ETL for Final Layer Workflow

Architecture Diagram:



The diagram above is the continuation of the Glue ETL with DQDL workflow, focusing on the passed silver data. After running the crawler to infer the schema of the passed data. Another Glue Job Script is used to perform data normalization and creation of fact and dim tables and lastly, partitioning the fact table to `order_purchase_date_key`. Once done, iceberg automatically creates a data catalog. AWS Athena will be used for ad-hoc queries to double check and explore the final layer dataset created.

Data Model - Star Schema



The Gold layer medallion is organized by using the star schema model. Quantitative transactional measurements are placed in the fact table. This fact table is referenced by the dimensional tables using foreign keys, enabling a more organized and efficient filtering of the data.